

Chapter 3 Determinants

Section 3.1: Introduction to Determinants

Section 3.2: Properties of Determinants

Section 3.3: Cramer's rule, Volume

Introduction to Determinants

- **Definition:** Let A be a 2×2 matrix, $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. The **determinant** of A is given by,

$$\det A = a_{11}a_{22} - a_{12}a_{21}$$

- **Definition:** The matrix A_{ij} is formed from the matrix A by removing the i -th row and j -th column of A .

Example If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, then $A_{13} = \begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix}$, and $A_{22} = \begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix}$

Introduction to Determinants

- **Definition:** For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is given by

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \det A_{1n}$$

$$= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j}$$

Introduction to Determinants

- **Example:** Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

- **Solution** Compute

$$\det A = a_{11}\det A_{11} - a_{12}\det A_{12} + a_{13}\det A_{13}:$$

$$= 1 \det \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix} - 5 \det \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} + 0 \det \begin{bmatrix} 2 & 4 \\ 0 & -2 \end{bmatrix}$$

$$= 1(0 - 2) - 5(0 - 0) + 0(-4 - 0) = -2$$

Introduction to Determinants

- Another common notation for the determinant of a matrix uses a pair of vertical lines in place of brackets.
- Thus the calculation in Example 1 can be written as

$$\det A = 1 \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5 \begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix} = \dots = -2$$

- To state the next theorem, it is convenient to write the definition of $\det A$ in a slightly different form. Given $A = [a_{ij}]$, the **(i, j) -cofactor** of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

Introduction to Determinants

- Then $\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$
- This formula is called a **cofactor expansion across the first row** of A .
- **Theorem:** The determinant of an $n \times n$ matrix A can be computed by a cofactor across any row or down any column.
- The expansion across the i th row using the cofactors is
$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$
- The cofactor expansion down the j th column is
$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

Introduction to Determinants

- **Example:** Use a cofactor expansion across the third row to compute $\det A$, where

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

- **Solution** Compute $\det A$

$$= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$$

$$= (-1)^{3+1}a_{31}\det A_{31} + (-1)^{3+2}a_{32}\det A_{32} + (-1)^{3+3}a_{33}\det A_{33}$$

$$= 0 \begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2) \begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix}$$

$$= 0 + 2(-1) + 0 = -2$$

Introduction to Determinants

- **Theorem:** If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .
- **Example:** Use a cofactor down the first column to compute $\det A$, where

$$A = \begin{bmatrix} 3 & 5 & 0 \\ 0 & -4 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

- **Solution** Compute $\det A$
$$= a_{11}C_{11} + 0C_{21} + 0C_{31}$$
$$= (-1)^{1+1}(3) \begin{vmatrix} -4 & -1 \\ 0 & 2 \end{vmatrix}$$
$$= (3)(-4)(2) = -24$$

Section 3.1: Introduction to Determinants

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Properties of Determinants

- **Theorem:** Let A be a square matrix
 - a) If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
 - b) If two rows of A are interchanged to produce B , then $\det B = -\det A$.
 - c) If one row of A is multiplied by k to produce B , then $\det B = k \det A$.

Properties of Determinants

- **Example:** Compute $\det A$, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$

- **Solution**

$$\det A = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

$$= - \begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

Properties of Determinants

- **Theorem:** A square matrix A is invertible if and only if $\det A \neq 0$.

- **Example** Compute $\det A$, where $A = \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$

- **Solution**

$$\det A = \begin{vmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{vmatrix} = \begin{vmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 0 & 0 & 0 \\ -5 & -8 & 0 & 9 \end{vmatrix} = 0$$

Column Operations

- **Theorem:** If A is a $n \times n$ matrix, then $\det A^T = \det A$.
- **Proof:** The theorem is obvious for $n = 1$. Suppose the theorem is true for $k \times k$ determinants and let $n = k + 1$.

Then the cofactor of a_{1j} in A equals the cofactor of a_{j1} in A^T , because the cofactors involve $k \times k$ determinants.

Hence the cofactor expansion of $\det A$ along the first row equals the cofactor expansion of $\det A^T$ down the first column. That is, A and A^T have equal determinants.

Thus the theorem is true for $n = 1$, and the truth of the theorem for one value of n implies its truth for the next value of n . By the principle of induction, the theorem is true for all $n \geq 1$.

Determinants and Matrix Products

- **Theorem:** If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.
- **Example** Verify Theorem 6 for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$.

Solution

$$AB = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$$

So $\det AB = 25(13) - 20(14) = 325 - 280 = 45$

Since $\det A = 9$ and $\det B = 5$,

$$(\det A)(\det B) = 9(5) = 45 = \det AB$$

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Cramer's rule

- **Theorem:** Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

where $A_i(\mathbf{b})$ is formed by replacing column i of A by $\mathbf{b}$.

- **Proof** Denote the columns of A by $\mathbf{a}_1, \dots, \mathbf{a}_n$ and the columns of the $n \times n$ identity matrix I by $\mathbf{e}_1, \dots, \mathbf{e}_n$.

Cramer's rule

- If $A\mathbf{x} = \mathbf{b}$, the definition of matrix multiplication shows that
$$A I_i(\mathbf{x}) = A[\mathbf{e}_1 \ \dots \ \mathbf{x} \ \dots \ \mathbf{e}_n] = [A\mathbf{e}_1 \ \dots \ A\mathbf{x} \ \dots \ A\mathbf{e}_n]$$
$$= [\mathbf{a}_1 \ \dots \ \mathbf{b} \ \dots \ \mathbf{a}_n] = A_i(\mathbf{b})$$
- By the multiplicative property of determinants,
$$(\det A) (\det I_i(\mathbf{x})) = \det A_i(\mathbf{b})$$
- The second determinant on the left is simply x_i . Hence $(\det A)x_i = \det A_i(\mathbf{b})$. This proves (1) because A is invertible and $\det A \neq 0$.

Cramer's rule

- **Example:** Use Cramer's rule to solve the system

$$\begin{aligned} 3x_1 - 2x_2 &= 6 \\ -5x_1 + 4x_2 &= 8 \end{aligned}$$

- **Solution** View the system as $A\mathbf{x} = \mathbf{b}$. Using the notation introduced above, $A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}$,

$$A_1(\mathbf{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(\mathbf{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

- Since $\det A = 2$, the system has a unique solution.

Cramer's rule

- By Cramer's rule,

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{24 + 16}{2} = 20$$

$$x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = \frac{24 + 30}{2} = 27$$

A Formula for A^{-1}

- The adjugate matrix is the *transpose* of the matrix of cofactors.

- **Theorem:** Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

A Formula for A^{-1}

- **Example:** Find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$.

- **Solution** The nine cofactors are

$$\begin{aligned} C_{11} &= + \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2, & C_{12} &= - \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3, & C_{13} &= + \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5 \\ C_{21} &= - \begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14, & C_{22} &= + \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, & C_{23} &= - \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7 \\ C_{31} &= + \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4, & C_{32} &= - \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1, & C_{33} &= + \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3 \end{aligned}$$

Thus

$$\text{adj } A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

A Formula for A^{-1}

- We could compute $\det A$ directly, but the following computation provides a check on the calculations above and produces $\det A$:

$$\begin{aligned} (\operatorname{adj} A) A &= \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{bmatrix} = 14 I \end{aligned}$$

A Formula for A^{-1}

- Since $(\text{adj } A) A = 14 I$, Theorem 8 shows that $\det A = 14$ and

$$A^{-1} = \frac{1}{14} \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} = \begin{bmatrix} -1/7 & 1 & 2/7 \\ 3/14 & -1/2 & 1/14 \\ 5/14 & -1/2 & -3/14 \end{bmatrix}$$

Determinants as Area or Volume

- **Theorem:** If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$.
- If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.
- **Proof** The theorem is obviously true for any 2×2 diagonal matrix:

$$\left| \det \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \right| = |ad| = \begin{cases} \text{area of} \\ \text{rectangle} \end{cases}$$

Determinants as Area or Volume

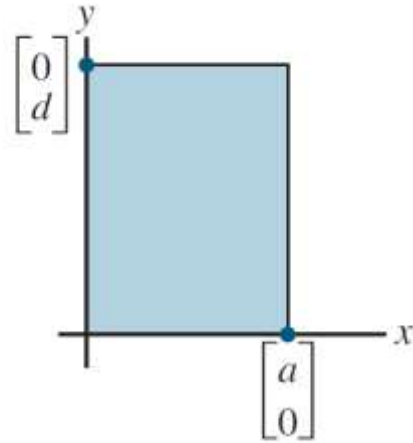


FIGURE 1

$$\text{Area} = |ad|.$$

- It will suffice to show that any 2×2 matrix $A = [\mathbf{a}_1 \ \mathbf{a}_2]$ can be transformed into a diagonal matrix in a way that changes neither the area of the associated parallelogram nor $|\det A|$.

Determinants as Area or Volume

- It suffices to prove the following simple geometric observation that applies to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$:
- Let $\mathbf{a}_1$ and $\mathbf{a}_2$ be nonzero vectors. Then for any scalar c , the area of the parallelogram determined by $\mathbf{a}_1$ and $\mathbf{a}_2$ equals the area of the parallelogram determined by $\mathbf{a}_1$ and $\mathbf{a}_2 + c\mathbf{a}_1$.
- To prove this statement, we may assume that $\mathbf{a}_2$ is not a multiple of $\mathbf{a}_1$, for otherwise the two parallelograms would be degenerate and have zero area.
- If L is the line through $\mathbf{0}$ and $\mathbf{a}_1$, then $\mathbf{a}_2 + L$ is the line through $\mathbf{a}_2$ parallel to L , and $\mathbf{a}_2 + c\mathbf{a}_1$ is on this line.

Determinants as Area or Volume

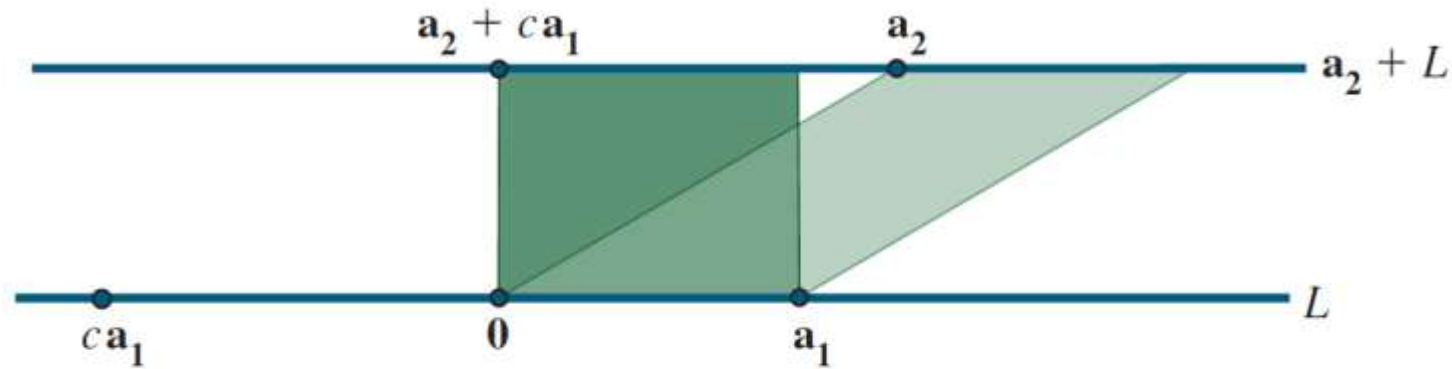


FIGURE 2 Two parallelograms of equal area.

- The points a_2 and $a_2 + ca_1$ have the same perpendicular distance to L . Hence the two parallelograms have the same area, since they share the base from 0 to a_1 .

Determinants as Area or Volume

- The proof for $\mathbb{R}^3$ is similar. The theorem is obviously true for a 3×3 diagonal matrix.

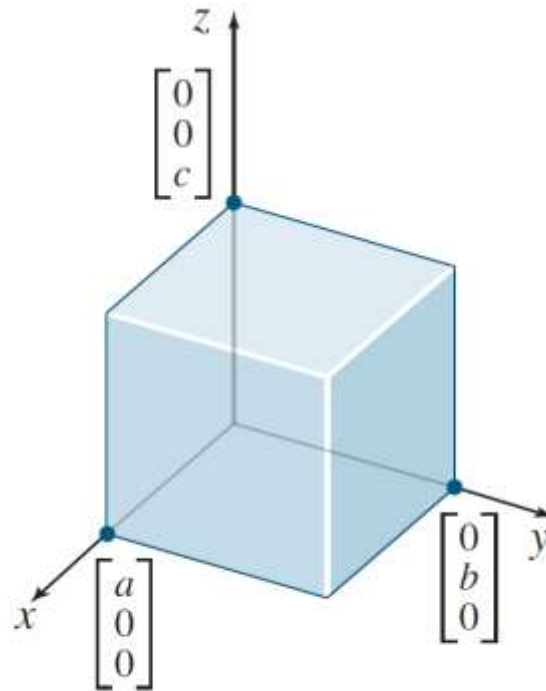


FIGURE 3

Volume = $|abc|$.

Determinants as Area or Volume

- And any 3×3 matrix A can be transformed into a diagonal matrix using column operations that do not change $|\det A|$.
- A parallelepiped is shown below as a shaded box with two sloping sides.

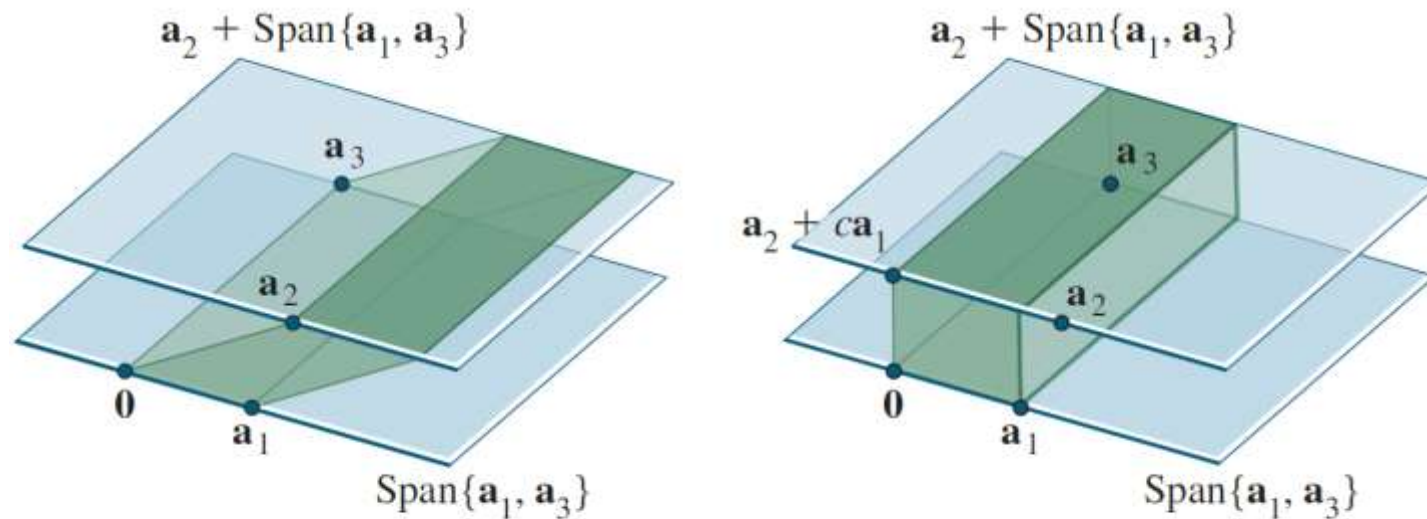


FIGURE 4 Two parallelepipeds of equal volume.

Determinants as Area or Volume

- Its volume is the area of the base in the plane $\text{Span}\{\mathbf{a}_1, \mathbf{a}_3\}$ times the altitude of $\mathbf{a}_2$ above $\text{Span}\{\mathbf{a}_1, \mathbf{a}_3\}$. Any vector $\mathbf{a}_2 + c\mathbf{a}_1$ lies in the plane $\text{Span}\{\mathbf{a}_1, \mathbf{a}_3\}$, which is parallel to $\text{Span}\{\mathbf{a}_1, \mathbf{a}_3\}$.
- Hence the volume of the parallelepiped is unchanged when $[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$ is changed to $[\mathbf{a}_1, \mathbf{a}_2 + c\mathbf{a}_1, \mathbf{a}_3]$.
- Thus a column replacement operation does not affect the volume of the parallelepiped. Since the column interchanges have no effect on the volume, the proof is complete.

Determinants as Area or Volume

- **Example 4** Calculate the area of the parallelogram determined by the points $(-2, -2)$, $(0, 3)$, $(4, -1)$, and $(6, 4)$.

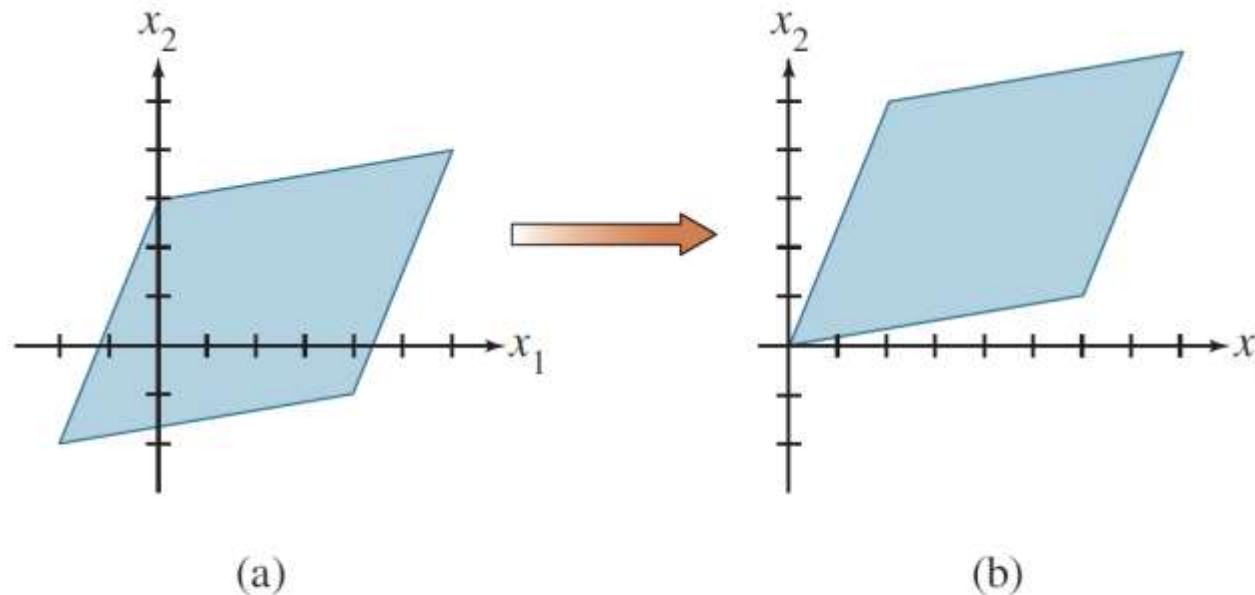


FIGURE 5 Translating a parallelogram does not change its area.

Determinants as Area or Volume

- **Solution** First translate the parallelogram to one having the origin as a vertex. For example, subtract the vertex $(-2, -2)$ from each of the four vertices.
- The new parallelogram has the same area, and its vertices are $(0, 0)$, $(2, 5)$, $(6, 1)$, and $(8, 6)$.
- This parallelogram is determined by the columns of
$$A = \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}$$
- Since $|\det A| = |-28|$, the area of the parallelogram is 28.